

* power supply

Volt	time	% Agarose
150 V	30 min	1%
120 V	1-1.5 h	2%
110 V	2-2.5 h	3%
9 V	3-3.5 h	4%

[N.A] → spectrophotometer.

Beer's - Lambert Law.

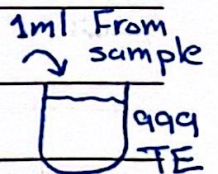
$$C \propto A$$

$$C = \epsilon * A$$

$$[ds DNA] = 50 \text{ Mg/ml} * A_{260}$$

$$[ss DNA] = 33 \text{ Mg/ml} * A_{260}$$

$$[RNA] = 40 \text{ Mg/ml} * A_{260}$$



↑
D.F = 1000 / D = 10³

$$D = 1/1000 = 10^{-3}$$

$$D.F = 1/D = 10^3$$

* UV → Aromatic (T, A, C, G)

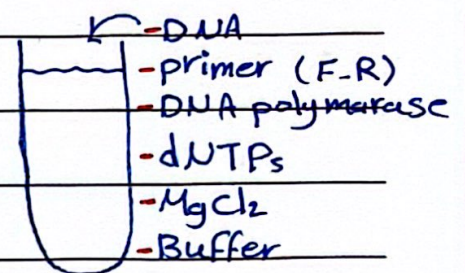
* The purity :-

- DNA = $A_{260} / A_{280} = (1.8 - 2)$ ^{RNA contamination}

- RNA = $A_{260} / A_{280} = (2.2 - 2.3)$ ^{protein contamination}

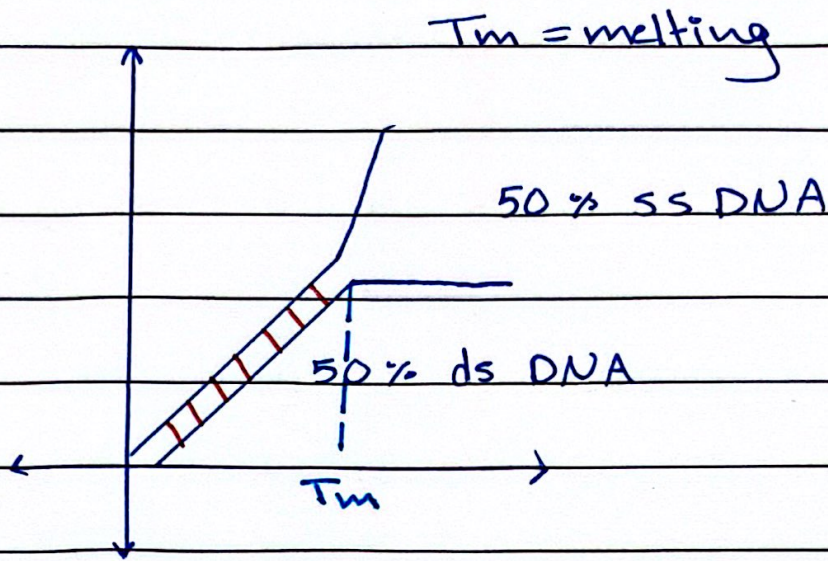
* primers

- ① 100% complementary
- ② Length 18-24
- ③ GC content (40-60%)
- ④ Difference between $T_m < 5^\circ$
- ⑤ Dimer (Homo - Hetero)



pcr Tube 25ml

* pcr - 3 cycle	T. C	Time	* of cycle
1- cycle 1	(85-97)°C	5-10 min	1
- initial denaturation			
2- cycle 2			30
1- Denaturation	(85-97)°C	30 sec	
2- Annealing	($T_m - 5$)°C		
3- Extension (Taq polymerase)	72°C	30 sec	
3- cycle 3	72°C	5-10 min	1
- Final Extension			
4- Holding (ایستایی)	4°C	∞	



$$T_m = 2(A+T) + 4(G+C)$$

$$T_m \cdot F = 2(10) + 4(10) = 60^\circ \text{C}$$

$$T_m \cdot R = 2(8) + 4(12) = 64^\circ \text{C}$$

← لقانونه الـ Annealing

$$\text{درجة الحرارة الأقل} \leftarrow T_m - 5 = 60 - 5 = 55^\circ \text{C}$$

$$\# \# \text{ of mol} = 2^n \rightarrow \# \text{ of cycle}$$

* Right → Revers
* Left → Forward

3' CTG ATC CAT GAT GGA CTC GAA 3'-OH
3' GAC TAG GTA CTA CCT GAG CTG ACT CCG AG ATCC ATG ACT AC AGG TC AGG CTC GAT AGG CA
3' CTG AATC CAT GAT GGA CTC GAC TGA GGC CTC TAGG TACT GAT GTC CAG TCC GAG CTA TCC GT
OH- 3' CAG GTC AGG CTC GAT AGG CA

F 5' CTG ATC CAT GAT GGA CTC GAA 3'
3' CAG GTC AGG CTC GAT GAT AGG CAA 3'
R 6 ~ 20 < 50%

* كينف بكون ال non-specific *

TACTAAGTTC
ATGACCTAG

ظا انك آخر وادوة رينف كينف ال primer